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(54) **TREE SKIRT USING SNAP-FIT STRUCTURE**

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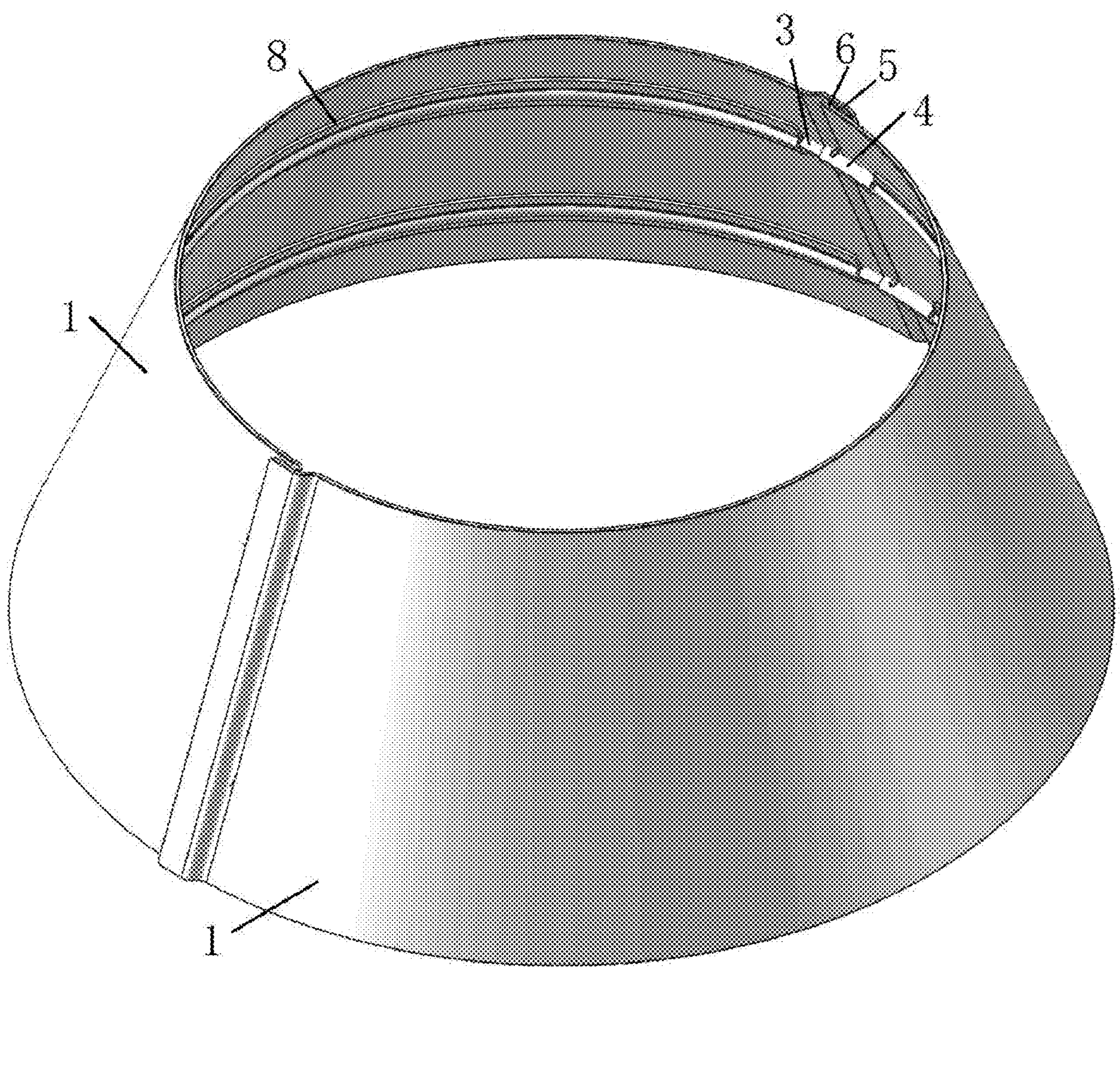
(57) **ABSTRACT**

A tree skirt using a snap-fit structure includes: a support assembly and at least two skirt pieces, the skirt pieces are sequentially arranged along a circular track, adjacent skirt pieces are detachably connected, and the skirt pieces are enclosed to form a cavity. The support assembly comprises at least two elastic strips, each of the elastic strips is arranged on an inner wall of the skirt piece, the elastic strips are sequentially arranged in an arrangement direction of the skirt pieces, one end of each of the elastic strips is provided with a snap member, the other end of the elastic strip is provided with a snap-fit member, the snap member on one of two adjacent elastic strips is snap-fixed with the snap-fit member on the other, and the snap member and the snap-fit member are detachably connected.



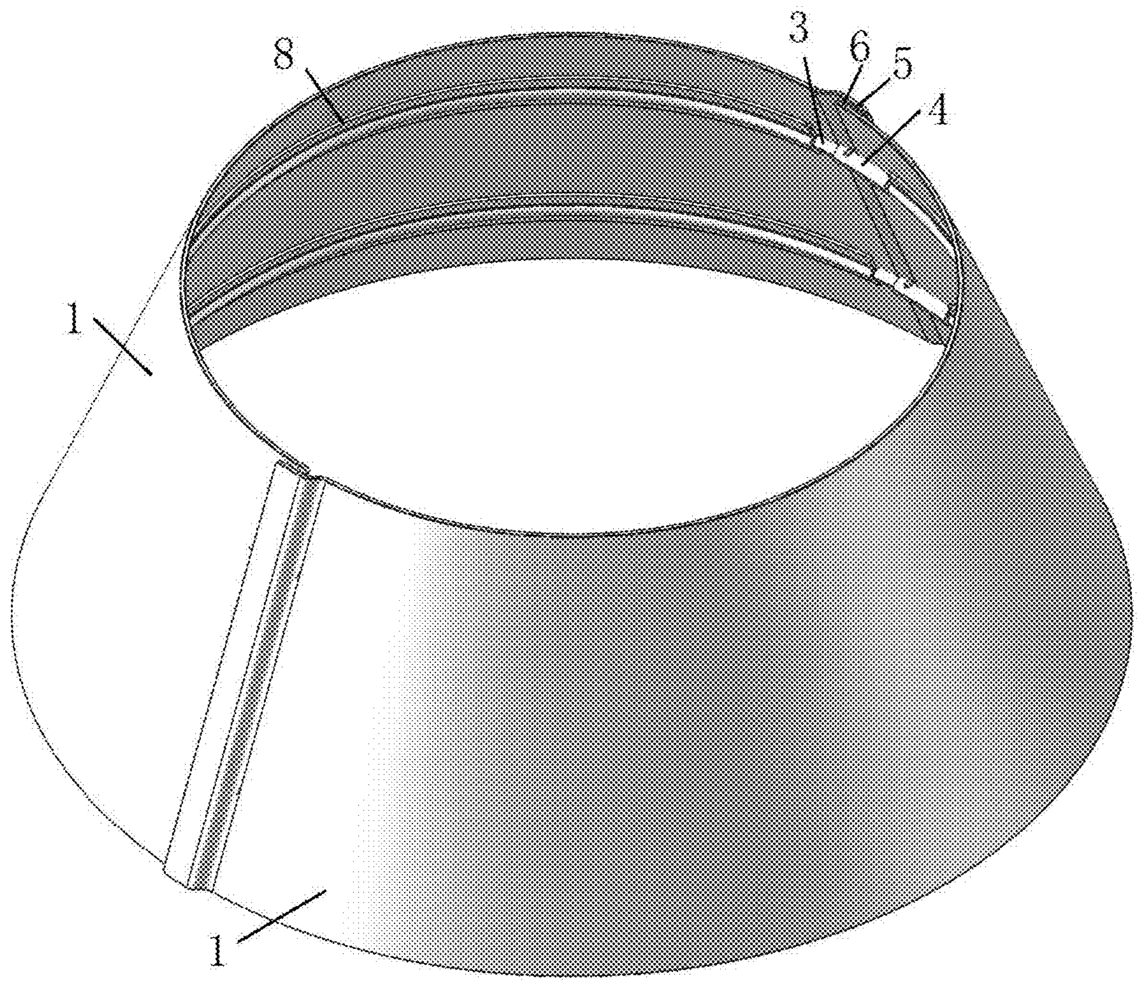


FIG. 1

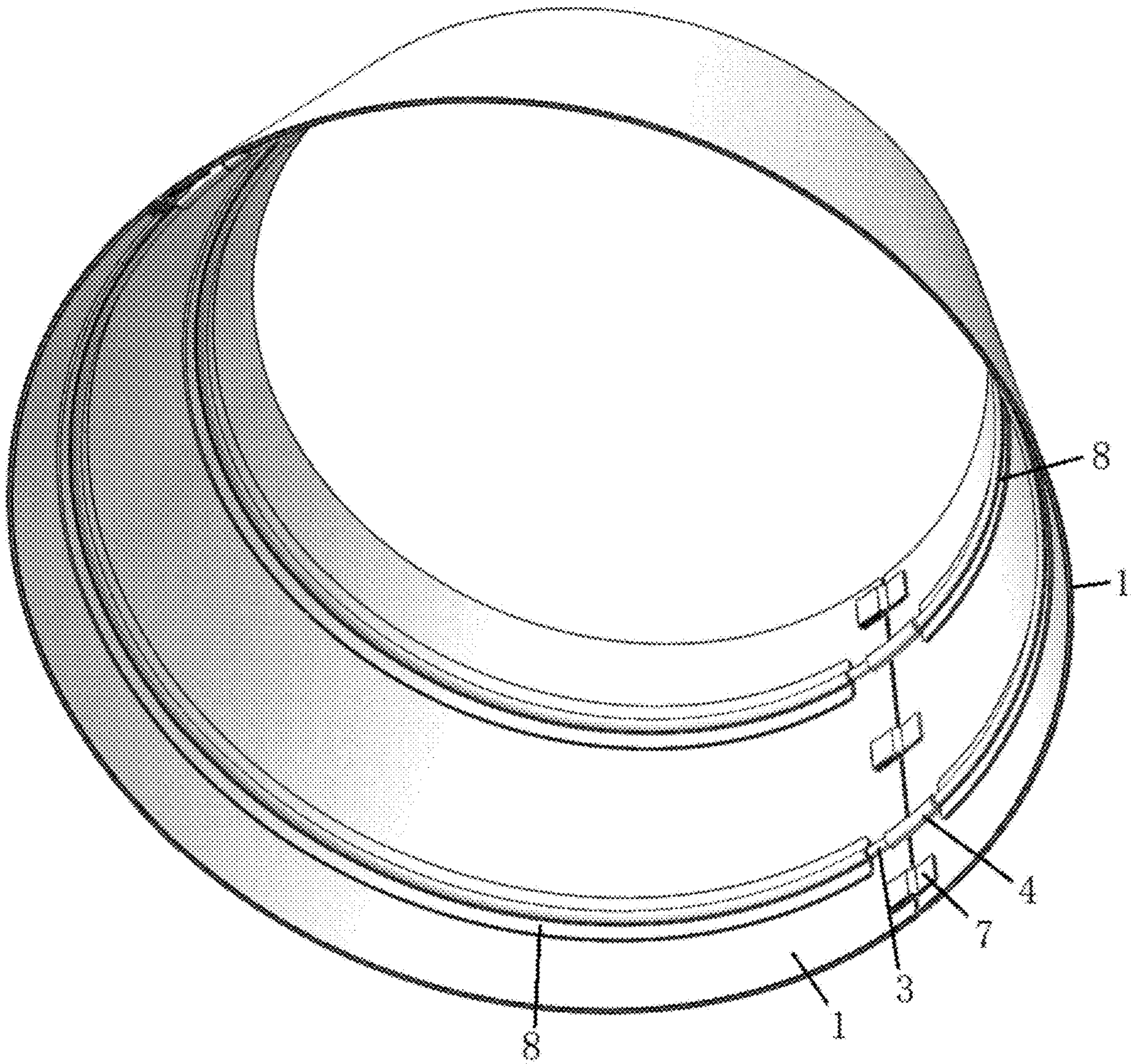


FIG. 2

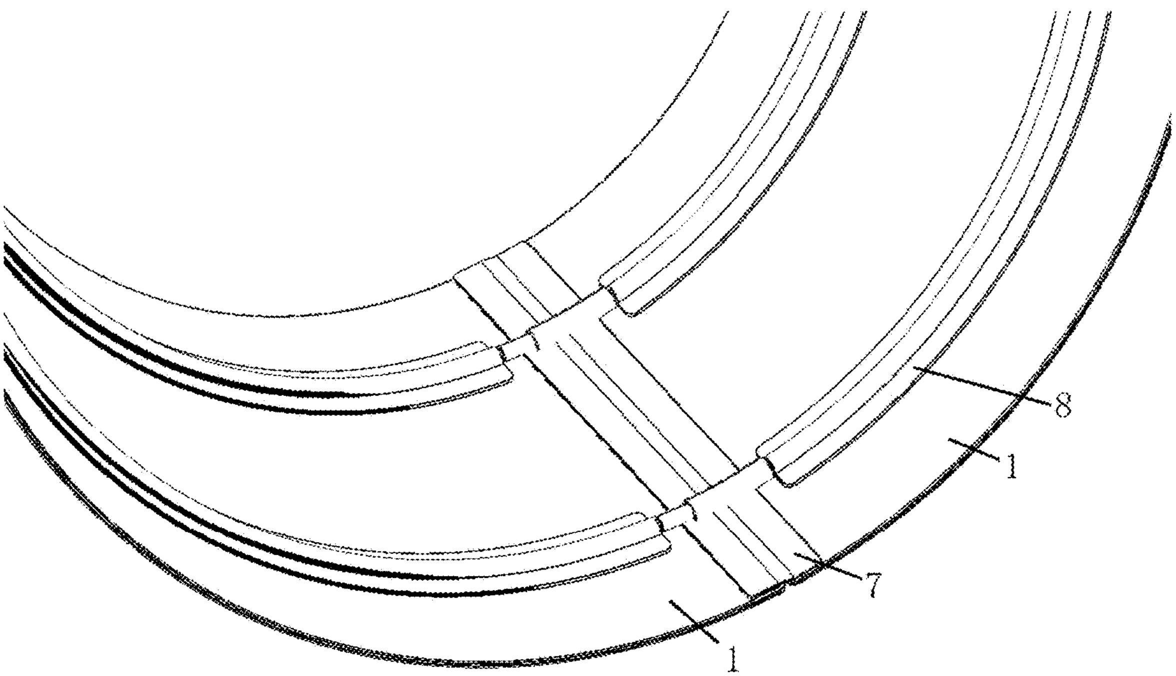


FIG. 3

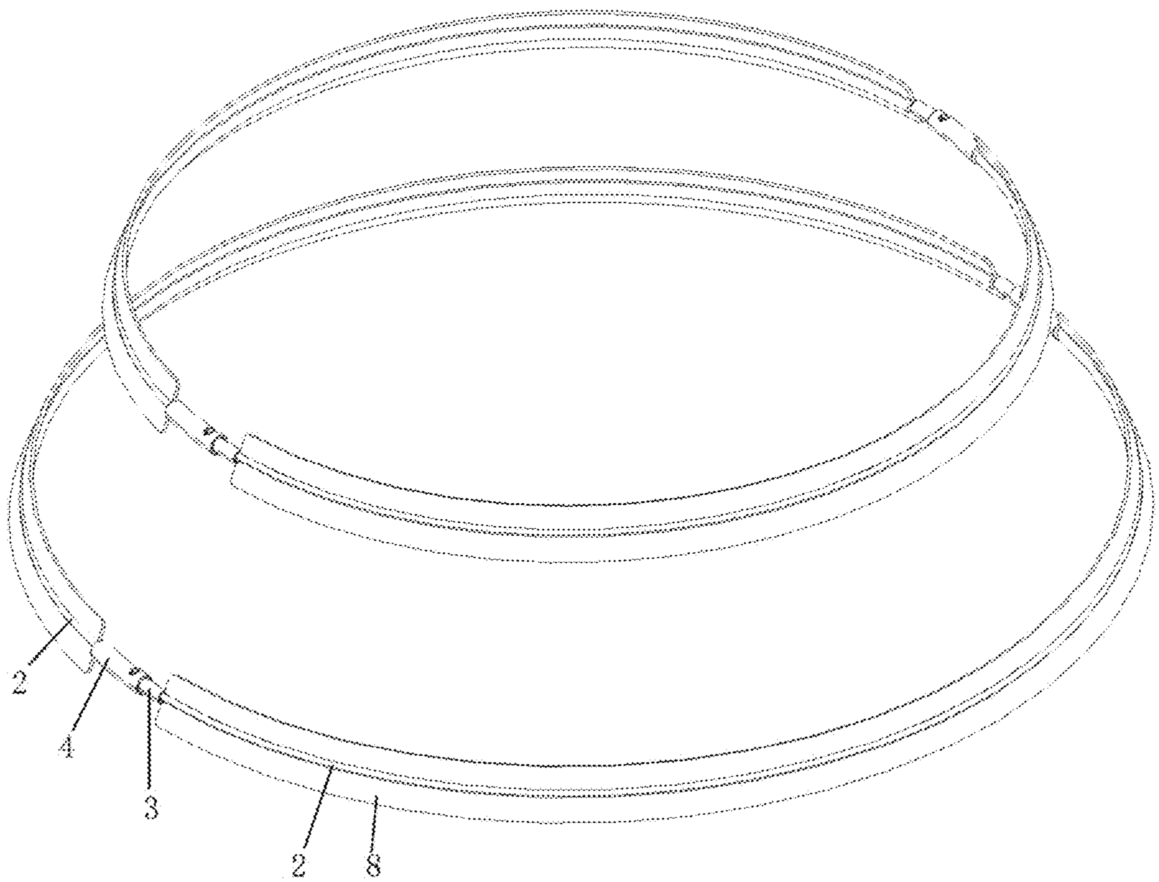


FIG. 4

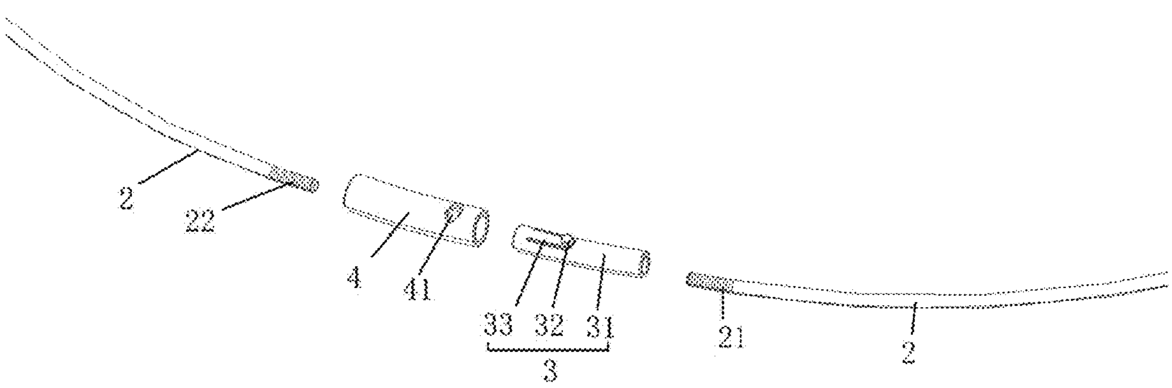


FIG. 5

TREE SKIRT USING SNAP-FIT STRUCTURE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202420338578.7, filed on Feb. 23, 2024, the content of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application pertains to the technical field of tree skirts, and specifically pertains to a tree skirt using a snap-fit structure.

BACKGROUND

[0003] Tree skirts are capable of decorating and protecting plants (e.g., plants such as Christmas trees that foil the festival ambience theme, daily household green plants, etc.). A tree skirt generally includes skirt pieces and a support assembly, the skirt pieces are enclosed to form a closed ring shape, and the skirt pieces extend around the plant. A support assembly generally includes a plurality of metal strips and a plurality of connecting members, each of the connecting members are located between adjacent metal strips, both ends of the connecting members have slots, and two adjacent metal strips are respectively inserted into corresponding slots. However, after the metal strips are inserted into the slots, the metal strips are mainly positioned by a friction force between the metal strips and inner walls of the slots, and the metal strips are prone to detachment from the connecting members after long-term use, resulting in deformation of the tree skirt, and leading to poor product stability, reliability and user experience.

[0004] In view of the problems, the present application is proposed.

SUMMARY

[0005] The present application provides a tree skirt using a snap-fit structure.

[0006] The present application provides the following technical solutions.

[0007] A tree skirt using a snap-fit configuration includes:

[0008] at least two skirt pieces, where the skirt pieces are sequentially arranged along a circular track, adjacent skirt pieces are detachably connected, and the skirt pieces are enclosed to form a cavity; and

[0009] a support assembly, where the support assembly includes at least two elastic strips, each of the elastic strips is arranged on an inner wall of the skirt piece, the elastic strips are sequentially arranged in an arrangement direction of the skirt pieces, one end of each of the elastic strips is provided with a snap member, the other end of the elastic strip is provided with a snap-fit member, the snap member on one of two adjacent elastic strips is snap-fixed with the snap-fit member on the other, and the snap member and the snap-fit member are detachably connected.

[0010] Optionally, the snap member includes an insertion body and a snap protrusion connected to the insertion body;

[0011] the snap-fit member has a slot and a snap hole communicating with the slot; and

[0012] in a state where the snap member and the snap-fit member are in a snap fit, the insertion body is inserted into the slot, and the snap protrusion is snapped into the snap hole.

[0013] Optionally, the snap protrusion is elastically movable in a diameter direction of the insertion body; and

[0014] during a process of inserting the insertion body into the slot, the snap protrusion moves towards an inside of the insertion body under the action of an inner wall of the slot, and when the insertion body moves to a point where the snap protrusion faces the snap hole, the snap protrusion moves away from the insertion body to be snapped into the snap hole.

[0015] Optionally, the insertion body has a cavity and a gap communicating with the cavity;

[0016] the snap protrusion is located in the gap, and the snap protrusion is connected to the insertion body by a connecting piece;

[0017] during the process of inserting the insertion body into the slot, the snap protrusion abuts against the inner wall of the slot, and the connecting piece is deformed towards an inside of the cavity of the insertion body, and when the insertion body moves to the point where the snap protrusion faces the snap hole, the connecting piece recovers deformation, driving the snap protrusion to be snapped into the snap hole.

[0018] Optionally, the connecting piece extends in a length direction of the insertion body; and

[0019] a free end of the connecting piece is provided with the snap protrusion.

[0020] Optionally, the snap hole is a circular hole; and

[0021] the snap protrusion is a cylinder.

[0022] Optionally, the elastic strip is inserted into the cavity of the insertion body, and the elastic strip is fixedly connected to the insertion body.

[0023] Optionally, an outer surface of one end of the elastic strip has a first concavo-convex pattern portion, so that the first concavo-convex pattern portion is in a close fit with an inner wall of the cavity in a state of being inserted into the cavity.

[0024] Optionally, an outer surface of one end of the elastic strip has a second concavo-convex pattern portion, so that the second concavo-convex pattern portion is in a close fit with the inner wall of the slot in a state where the elastic strip is inserted into the slot of the snap-fit member.

[0025] Optionally, the tree skirt includes two support assemblies, the skirt pieces are enclosed to form an upper opening and a lower opening, and the two support assemblies are sequentially arranged in a direction from the lower opening to the upper opening.

[0026] By adopting the above technical solutions, the present application has the following beneficial effects.

[0027] The elastic strips of the tree skirt of the present application are connected and fixed by a snap-in structure, which results in a stable connection structure, therefore, the adjacent elastic strips are not easy to be detached, leading to good stability and high reliability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The accompanying drawings, which are incorporated in and constitute a part of the present application, are included to provide a further understanding of the present application. Illustrative embodiments of the present application and the description thereof are included to illustrate the present application, but are not to be construed as unduly limiting the present application. Apparently, the accompanying drawings in the following description are merely some rather than all embodiments of the present application, and a person of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts.

[0029] FIG. **1** illustrates a first tree skirt using a snap-fit structure according to an embodiment of the present application.

[0030] FIG. **2** illustrates a second tree skirt using a snap-fit structure according to an embodiment of the present application.

[0031] FIG. **3** is an inner view of a third tree skirt using a snap-fit structure according to an embodiment of the present application.

[0032] FIG. **4** is a structural diagram of a tree skirt using a snap-fit structure with skirt pieces removed according to an embodiment of the present application.

[0033] FIG. **5** is an exploded view of a fit structure of an elastic strip, a snap member and a snap-fit member of a tree skirt using a snap-fit structure according to an embodiment of the present application.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0034] In order to make the purpose, technical solutions and advantages of the embodiments of the present application more clear, the technical solutions in the embodiments of the present application will be clearly and completely described below in conjunction with the accompanying drawings in the embodiments of the present application. The embodiments described below are for illustrative purposes only and are not intended to limit the scope of the present application.

[0035] In the description of the present application, it should be noted that the terms "upper", "lower", "inner", "outer" and the like indicate orientations or positional relationships based on the orientations or positional relationships shown in the drawings, and are only for convenience of description and simplification of description, but do not indicate or imply that the device or element to be referred must have a specific orientation, be constructed and be operated in a specific orientation, and thus, should not be construed as limiting the present application.

[0036] In the description of the present application, it should be noted that, unless expressly and specifically defined otherwise, the terms "mounted" and "connected" are to be interpreted broadly, and may, for example, be fixedly or detachably or integrally connected; may be mechanically connected or electrically connected; or may be directly connected or indirectly connected through an intermediary. The specific meaning of the above terms in the present application can be understood by those of ordinary skill in the art from the specific context.

[0037] As shown in FIGS. **1** to **5**, an embodiment of the present application provides a tree skirt using a snap-fit structure, including: a support assembly and at least two skirt pieces **1**, the skirt pieces **1** are sequentially arranged along a circular track, adjacent skirt pieces **1** are detachably connected, and the skirt pieces **1** are enclosed to form a cavity. The support assembly includes at least two elastic strips **2**, each of the elastic strips **2** is arranged on an inner wall of the skirt piece **1**, the elastic strips **2** are sequentially arranged in an arrangement direction of the skirt pieces **1**, one end of each of the elastic strips **2** is provided with a snap member **3**, the other end of the elastic strip **2** is provided with a snap-fit member **4**, the snap member **3** on one of two adjacent elastic strips **2** is snap-fixed with the snap-fit member **4** on the other, and the snap member **3** and the snap-fit member **4** are detachably connected. The elastic strips **2** of the tree skirt of the present application are connected and fixed by a snap-in structure, which results in a stable connection structure, therefore, the adjacent elastic strips **2** are not easy to be detached, leading to good stability and high reliability.

[0038] In some possible embodiments, as shown in FIG. **5**, the snap member **3** includes an insertion body **31** and a snap protrusion **32** connected to the insertion body, the snap-fit member **4** has a slot and a snap hole **41** communicating with the slot, and in a state where the snap member **3** and the snap-fit member **4** are in a snap fit, the insertion body **31** is inserted into the slot, and the snap protrusion **32** is snapped into the snap hole **41**.

[0039] For example, the snap protrusion **32** is elastically movable in a diameter direction of the insertion body **31**; and during a process of inserting the insertion body **31** into the slot, the snap protrusion **32** moves towards an inside of the insertion body **31** under the action of an inner wall of the slot, and when the insertion body **31** moves to a point where the snap protrusion **32** faces the snap hole **41**, the snap protrusion **32** moves away from the insertion body **31** to be snapped into the snap hole **41**.

[0040] Specifically, the insertion body **31** has a cavity and a gap communicating with the cavity, the snap protrusion **32** is located in the gap, and the snap protrusion **32** is connected to the insertion body **31** by a connecting piece **33**. During the process of inserting the insertion body **31** into the slot, the snap protrusion **32** abuts against the inner wall of the slot, and the connecting piece **33** is deformed towards an inside of the cavity of the insertion body **31**, and when the insertion body **31** moves to the point where the snap protrusion **32** faces the snap hole **41**, the connecting piece **33** recovers deformation, driving the snap protrusion **32** to be snapped into the snap hole **41**. The snap member **3** and the snap-fit member **4** may be metal members or plastic members.

[0041] In some possible embodiments, the connecting piece **33** extends in a length direction of the insertion body **31**, and a free end of the connecting piece **33** is provided with the snap protrusion **32**. The connecting piece **33** has a long length and a certain elasticity, so that the snap protrusion **32** may be smoothly snapped into the snap hole **41** of the snap-fit member **4**.

[0042] In some possible embodiments, the snap hole **41** is a circular hole, and the snap protrusion **32** is a cylinder. The snap protrusion **32** is a cylinder which may be smoothly inserted into the circular hole. When it is necessary to disassemble the snap member **3** and the snap-fit member **4**, the snap protrusion **32** is pressed so that the snap protrusion **32** is retracted into the cavity, and then an opposite force is applied to the snap member **3** and the snap-fit member **4** respectively, so that the snap member **3** is detached from the snap-fit member **4**.

[0043] In some possible embodiments, as shown in FIG. **5**, the elastic strip **2** is inserted into the cavity of the insertion body **31**, and the elastic strip **2** is fixedly connected to the insertion body **31**. The elastic strip **2** may be welded to the insertion body **31**, and the elastic strip **2** may also be in an interference fit with the insertion body **31**, and the two are connected to form an integral structure.

[0044] In some possible embodiments, an outer surface of one end of the elastic strip **2** has a first concavo-convex pattern portion **21**, so that the first concavo-convex pattern portion **21** is in a close fit with an inner wall of the cavity in a state of being inserted into the cavity. Arrangement of the concavo-convex pattern portion increases a frictional force between the elastic strip **2** and the inner wall of the insertion body **31**, and improves connection stability between the elastic strip **2** and the insertion body **31**.

[0045] In some possible embodiments, an outer surface of one end of one elastic strip **2** has a second concavo-convex pattern portion **22**, so that the second concavo-convex pattern portion **22** is in a close fit with the inner wall of the slot in a state where the elastic strip **2** is inserted into the slot of the snap-fit member. Arrangement of the concavo-convex pattern portions increases a frictional force between the elastic strip **2** and an inner wall of the snap-fit member, and improves connection stability between the elastic strip **2** and the snap-fit member.

[0046] In some possible embodiments, as shown in FIGS. **1** to **4**, the tree skirt includes two support assemblies, the skirt pieces **1** are enclosed to form an upper opening and a lower opening, and the two support assemblies are sequentially arranged in a direction from the lower opening to the upper opening. By arranging the two support assemblies, the skirt pieces **1** may be supported at both a position close to the upper opening and a position close to the lower opening, so that the skirt pieces **1** remain upright as a whole.

[0047] In some possible embodiments, as shown in FIG. **1**, one of adjacent two skirt pieces **1** is provided with a loop fastener **5**, and the other is provided with a hook fastener **6**. The loop fastener **5** on one of the adjacent two skirt pieces and the hook fastener **6** on the other are fixed by adhering.

[0048] As shown in FIG. **1**, the skirt piece **1** has main surfaces perpendicular to a thickness direction thereof, and both ends of the main surfaces are respectively provided with the loop fastener **5** and the hook fastener **6**. The main surfaces of the adjacent two skirt pieces **1** at the both ends are fit together, and the loop fastener **5** on the main surfaces of one of the adjacent two skirt pieces is adhered to the hook fastener **6** on the main surfaces of the other skirt piece. The skirt piece **1** has two main surfaces respectively located on both sides of the skirt piece **1** in the thickness direction. The loop fastener **5** and a second hook fastener **6** are respectively arranged on the two main surfaces.

[0049] In some possible embodiments, as shown in FIG. **3**, the both ends of the skirt piece are respectively provided with a width edge, a first adhering member is provided on the width edge of one of the two adjacent skirt pieces, and an extension piece **7** is provided on the width edge of the other, the extension piece **7** is provided with a second adhering member, thickness end faces of the two skirt pieces are fitted together, and the second adhering member on the extension piece is adhered to the first adhering member. The extension piece **7** is located inside the skirt piece, which does not affect appearance of the skirt piece. One of the first adhering member and the second adhering member may be a loop fastener, and the other may be a hook fastener.

[0050] In some possible embodiments, as shown in FIG. **2**, the both ends of the skirt piece **1** are respectively provided with a width edge, a plurality of first adhering members are provided on the width edge of one of the adjacent two skirt pieces **1**, the first adhering members are sequentially spaced apart along the width edge, and a plurality of extension pieces **7** are provided on the width edge of the other at intervals, and each of the extension pieces **7** is provided with a second adhering member. Thickness end surfaces of the adjacent two skirt pieces **1** are fitted together, and the second adhering members on each of the extension pieces **7** on a corresponding width edge of one of the adjacent two skirt pieces are respectively connected to corresponding first adhering members on the width edge of the other skirt piece. One of the first adhering member and the second adhering member may be a loop fastener, and the other may be a hook fastener.

[0051] The tree skirt further includes a connecting member **8**, both sides of the connecting member **8** are respectively connected to an inner surface of the skirt piece **1**, the connecting member **8** and the skirt piece **1** are enclosed to form a threading cavity, and the metal strip **2** is arranged through the threading cavity. The connecting member **8** may be adhered to the skirt piece **1** by a hook and loop fastener, the connecting member **8** may also be sewn to the skirt piece **1** by means of sutures, and the connecting member **8** may also be glued to the skirt piece **1** by means of glue.

[0052] The preferred embodiments of the present application disclosed above are intended only to illustrate the present application. It is not intended to elaborate all details of the preferred embodiments, nor is it intended to limit the application to only the specific embodiments described. Obviously, many modifications and variations are possible in light of the above teaching. The embodiments were chosen and described in order to best explain the principles and practical application of the present application and to thereby enable those skilled in the art to best understand and utilize the present application. The present application is to be limited only by the claims, along with their full scope and equivalents.

What is claimed is:

1. A tree skirt using a snap-fit structure, comprising:

at least two skirt pieces, wherein the skirt pieces are sequentially arranged along a circular track, adjacent skirt pieces are detachably connected, and the skirt pieces are enclosed to form a cavity; and

a support assembly, wherein the support assembly comprises at least two elastic strips, each of the elastic strips is arranged on an inner wall of the skirt piece, the elastic strips are sequentially arranged in an arrangement direction of the skirt pieces, one end of each of the elastic strips is provided with a snap member, the other end of the elastic strip is provided with a snap-fit member, the snap member on one of two adjacent elastic strips is snap-fixed with the snap-fit member on the other of the two adjacent elastic strips, and the snap member and the snap-fit member are detachably connected.

2. The tree skirt according to claim **1**, wherein the snap member comprises an insertion body and a snap protrusion connected to the insertion body;

the snap-fit member has a slot and a snap hole communicating with the slot; and

in a state where the snap member and the snap-fit member are in a snap fit, the insertion body is inserted into the slot, and the snap protrusion is snapped into the snap hole.

3. The tree skirt according to claim **2**, wherein the snap protrusion is elastically movable in a diameter direction of the insertion body; and

during a process of inserting the insertion body into the slot, the snap protrusion moves towards an inside of the insertion body under the action of an inner wall of the slot, and when the insertion body moves to a point where the snap protrusion faces the snap hole, the snap protrusion moves away from the insertion body to be snapped into the snap hole.

4. The tree skirt according to claim **3**, wherein the insertion body has a cavity and a gap communicating with the cavity;

the snap protrusion is located in the gap, and the snap protrusion is connected to the insertion body by a connecting piece; and

during the process of inserting the insertion body into the slot, the snap protrusion abuts against the inner wall of the slot, and the connecting piece is deformed towards an inside of the cavity of the insertion body, and when the insertion body moves to the point where the snap protrusion faces the snap hole, the connecting piece recovers deformation, driving the snap protrusion to be snapped into the snap hole.

5. The tree skirt according to claim **4**, wherein the connecting piece extends in a length direction of the insertion body; and

a free end of the connecting piece is provided with the snap protrusion.

6. The tree skirt according to claim **4**, wherein the snap hole is a circular hole; and

the snap protrusion is a cylinder.

7. The tree skirt according to claim **4**, wherein the elastic strip is inserted into the cavity of the insertion body, and the elastic strip is fixedly connected to the insertion body.

8. The tree skirt according to claim **7**, wherein an outer surface of one end of the elastic strip has a first concavo-convex pattern portion, so that the first concavo-convex pattern portion is in a close fit with an inner wall of the cavity in a state of being inserted into the cavity.

9. The tree skirt according to claim **2**, wherein an outer surface of one end of the elastic strip has a second concavo-convex pattern portion, so that the second concavo-convex pattern portion is in a close fit with the inner wall of the slot in a state where the elastic strip is inserted into the slot of the snap-fit member.

10. The tree skirt according to claim **1**, comprising two support assemblies, wherein the skirt pieces are enclosed to form an upper opening and a lower opening, and the two support assemblies are sequentially arranged in a direction from the lower opening to the upper opening.

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